

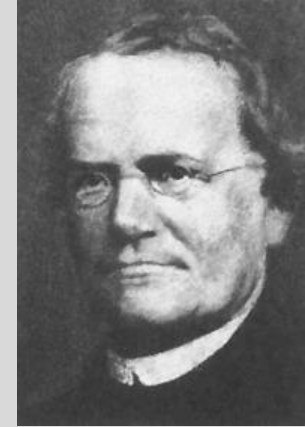
MOLECULAR GENETICS I: Chromosomal Structure and DNA Replication



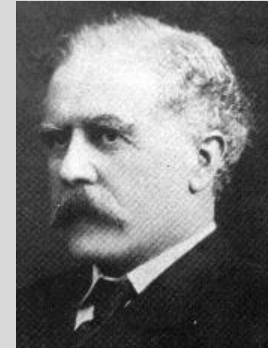
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Founding Director of Masters in Biomedical Sciences
Department of Biomedical and Anatomical Sciences
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1. State the significance of genetics in medicine
2. Outline key experiments that led to our understanding of genetics
3. Identify key aspects of DNA structure
4. Identify key aspects of chromosomal structure and organization
5. Identify the roles of major proteins involved in DNA replication
6. Compare and contrast differences between prokaryotic and eukaryotic replication

- Central component to our understanding of most major diseases
- All processes are influenced by genes
- Mendel (1822-1884)
- Archibald Garrod (1902) described alkaptonuria as the first “inborn error of metabolism.”
- Johanssen (1909) coined the term gene to denote the basic unit of heredity

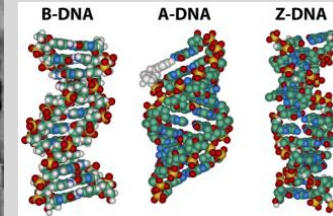
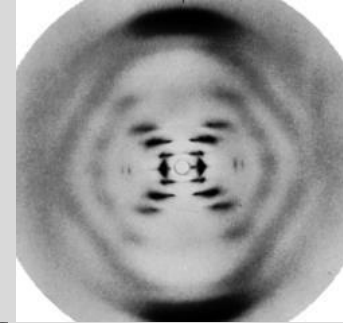


L. B. Jorde PhD, J. C. Carey MD, MPH and M. J. Bamshad MD. 2016. **Medical Genetics**. 5th Ed. Elsevier, Inc.

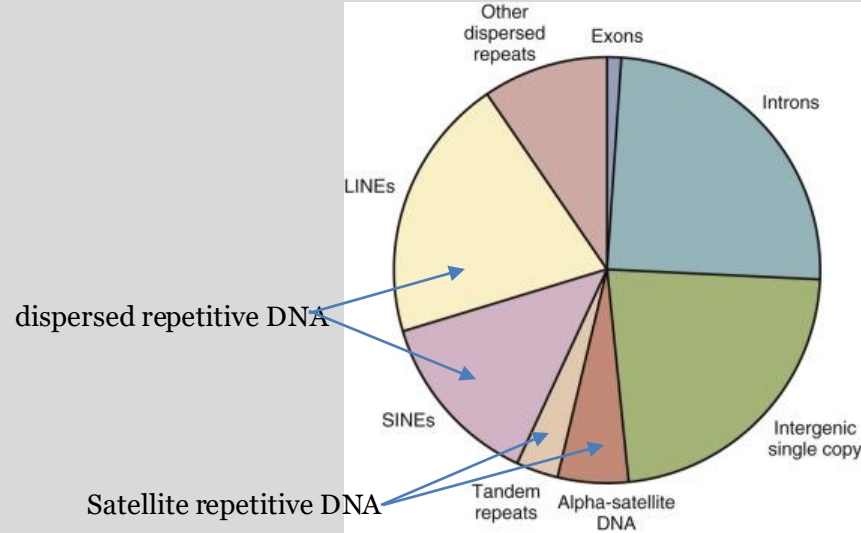


Sir Archibald Garrod,
around 1910.

- Avery–MacLeod–McCarty (1944) - genes are composed of **DNA**
- Watson & Crick (1953)
- Plasmids & restriction enzymes
- PCR & cloning
- Human Genome
- Post-genomic era

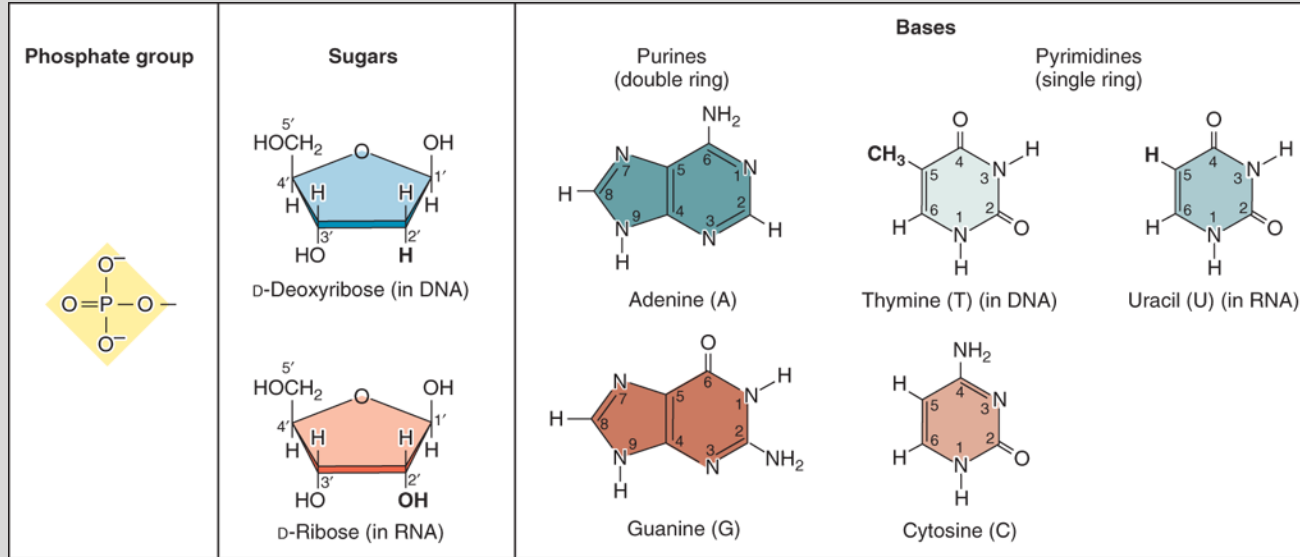


- 1-2% of the 6 billion bps protein coding

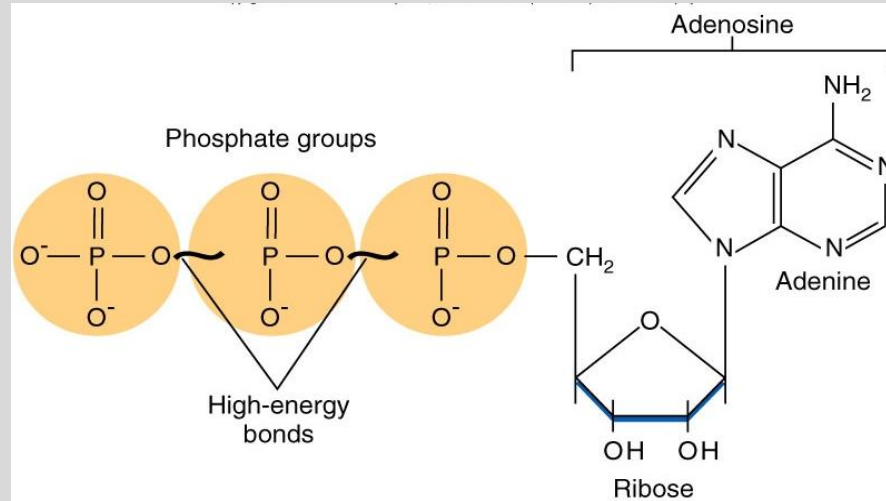


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Medical Genetics. 6th Ed. Elsevier, Inc.

Nucleic Acids Structure (nucleotide composition)

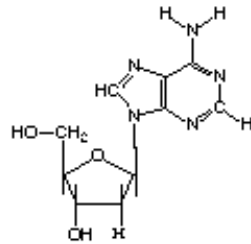


Source: G. Bradley Schaefer, James N. Thompson, Jr.:
Medical Genetics: An Integrated Approach
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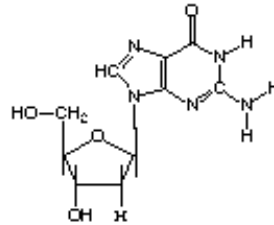


Source: The McGraw Hill Publishers

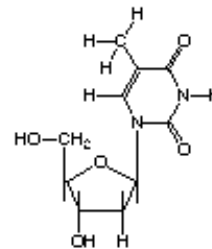
The Nucleotides of DNA



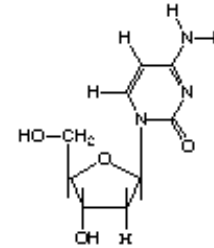
Adenine



Guanosine



Thymine

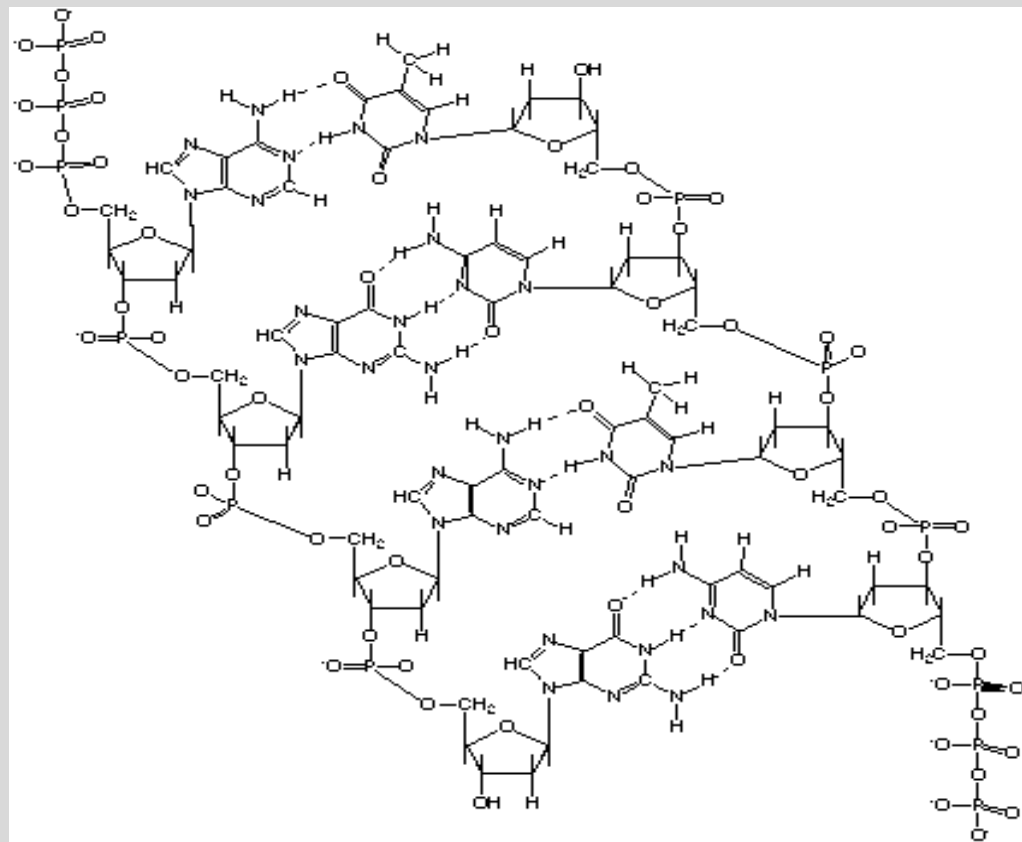


Cytosine

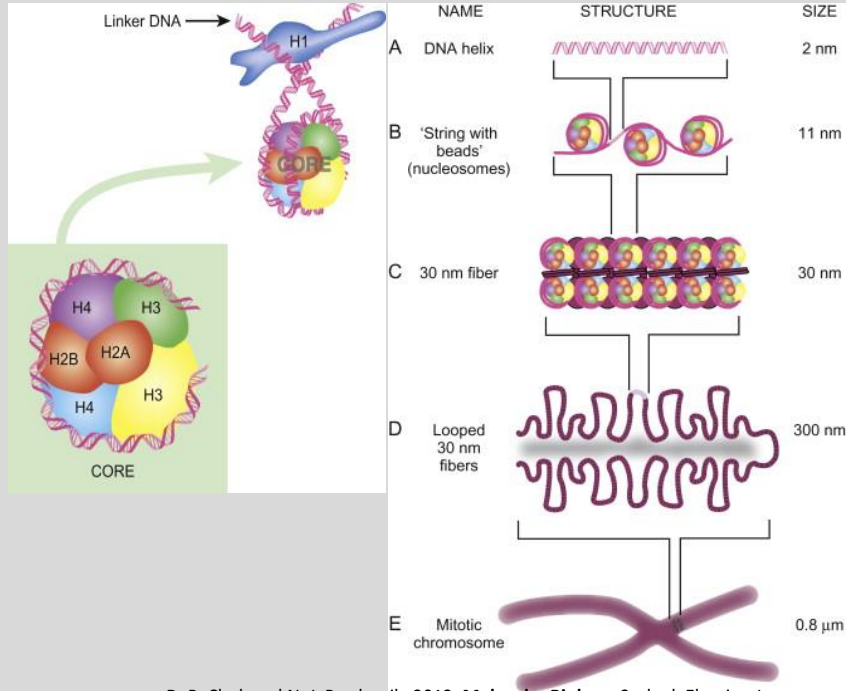
Purines

Pyrimidines

http://cs.boisestate.edu/~amit/teaching/342/lab/structure_files/DNAnucleotides.gif



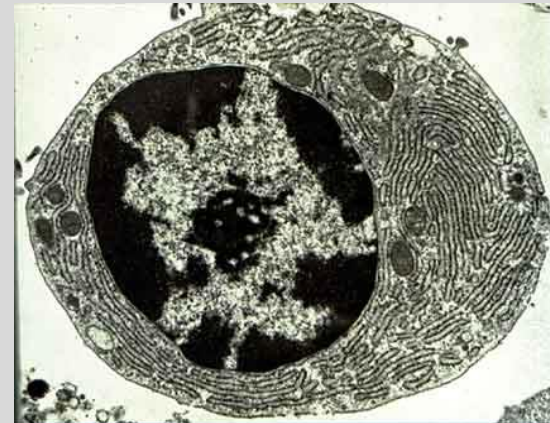
- Nucleosome/ Histones
- Heterochromatin/ Euchromatin



D. P. Clark and N. J. Pazdernik. 2013. **Molecular Biology**, 2nd ed. Elsevier, Inc.

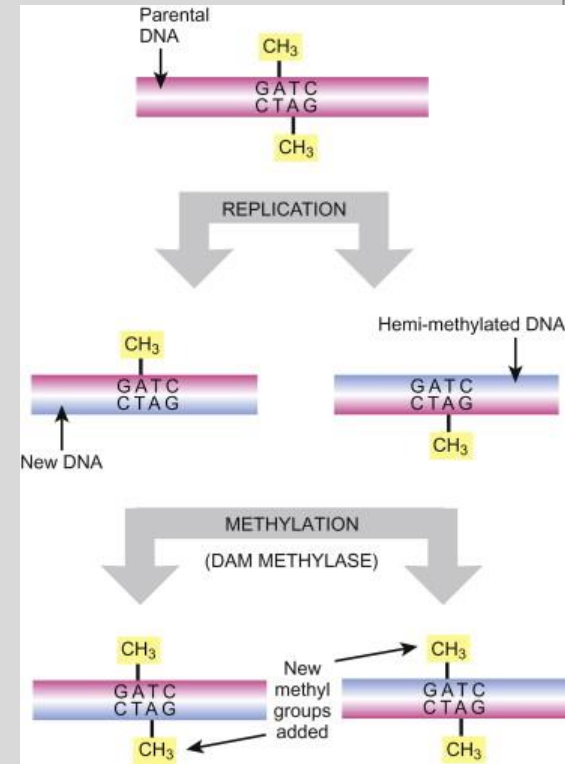


http://www.sciencephoto.com/image/209697/530wm/G1100198-Coloured_TEM_of_DNA_from_E._coli_bacterium-SPL.jpg

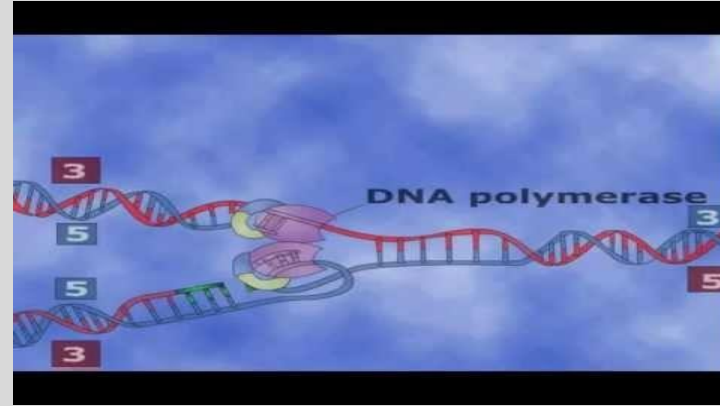


<https://theadventuresofbecky.files.wordpress.com/2010/03/plasmid-a-cell-em.jpg>

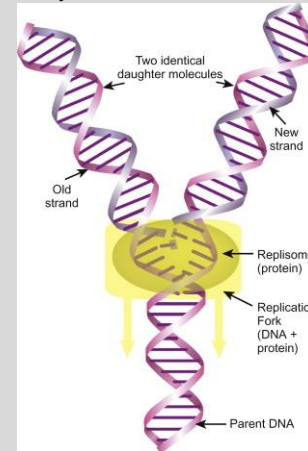
- DNA methylation:
 - a) GATC - mismatch repair
 - b) CpG islands - silencing
- Histone methylation/acetylation:
 - Methylation: reversibly repress transcription
 - Acetylation: relaxes DNA coiling allows for transcription



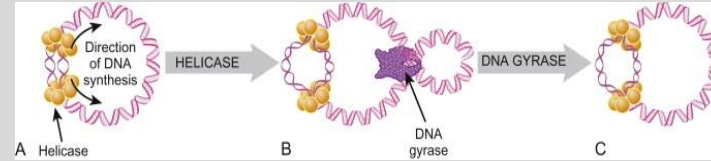
- Ori/ter
- DNA helicase, SSBPs, Primase, DNA Pol III, DNA Pol I, DNA ligase
- 5' → 3'
- Semi-conservative
- Pol III proof reading ability



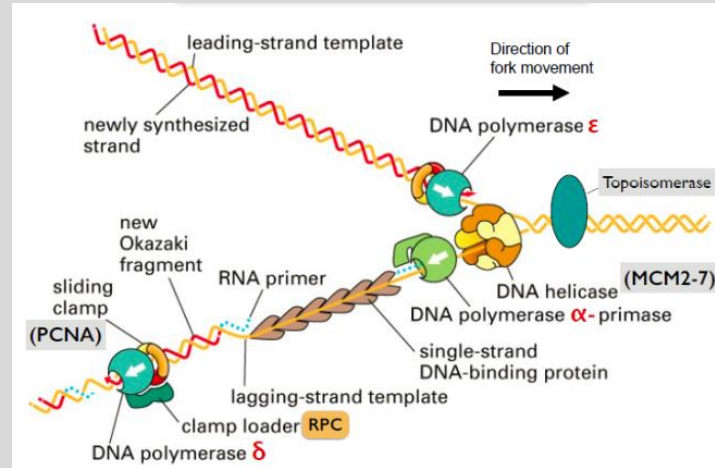
https://www.youtube.com/watch?v=OSRMht3_hHE



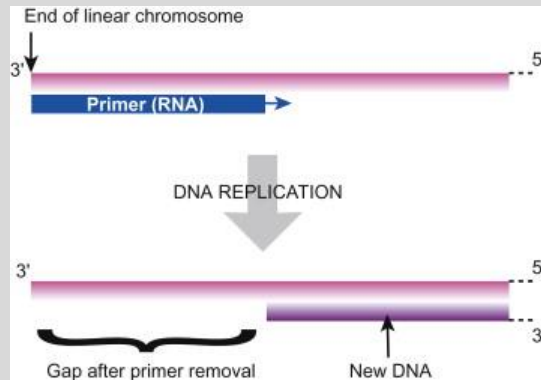
- Prokaryotic: DNA gyrase (nalidixic acid/ciprofloxacin)
- Eukaryotic: chromatin remodeling
- 3 pol: α (iDNA), ϵ (leading), & δ (lagging)



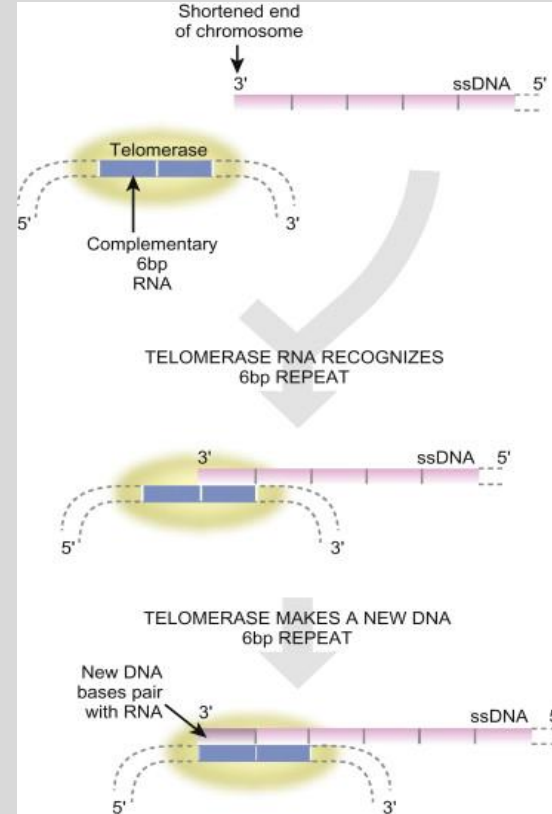
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- Telomeres: mainly expressed in germ cells



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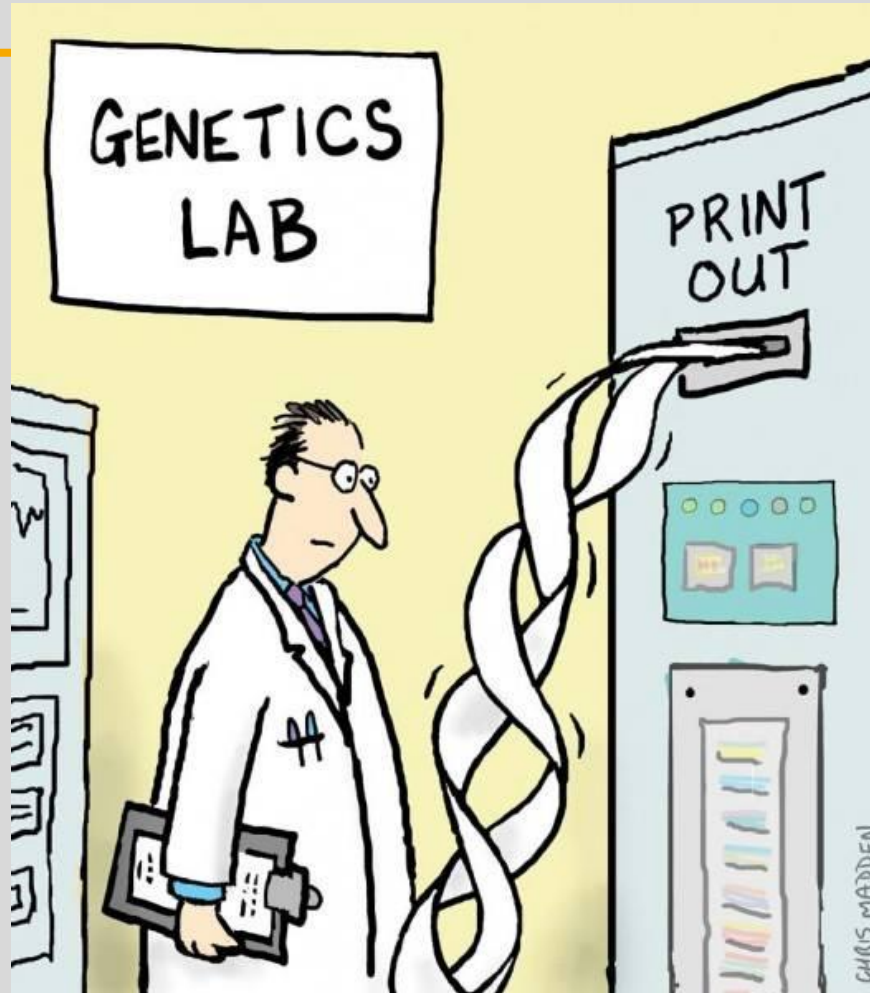


D. P. Clark and N. J. Pazdernik. 2013. **Molecular Biology**, 2nd ed. Elsevier, Inc.

- Brief history of genetics and significance
- Basic nucleic acid structure
- Chromatin structure/modifications
- DNA replication in prokaryotes and eukaryotes

Core References:

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